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Sustainable Housing Suitability Analysis

Austin-Round Rock-San Marcos MSA

Introduction

Over the past decade, Texas has consistently had one of the highest population growth rates in the United States. According to the 2020 census, the state reported a 15.9% ten-year population change, making it one of the fastest-growing states in the country (U.S. Census Bureau, 2021). For the third consecutive year, Texas added more residents than any other state, with its overall population growing by over 391,000 between July 2024 and July 2025 (U.S. Census Bureau, 2025).

Much of this growth is contained within a three-point polygon known as the "Texas Triangle," which spans from the Dallas-Fort Worth area in the north to San Antonio in the west and Houston in the southeast. The Texas Triangle is one of eleven distinct megaregions in the United States (Clark, 2021). An exploratory data analysis on population density conducted as part of this project found the edges and vertices of this triangle to be the densest areas in the state (Figure 1).

The Austin–Round Rock–San Marcos Metropolitan Statistical Area (MSA) sits at the western apex of the Texas Triangle and contains some of the most consistently fastest-growing cities in the state, including Austin, Round Rock, and Pflugerville. This sustained growth has put enormous pressure on the region's housing supply. A 2023 report from the Austin Board of Realtors found that the Austin metro came up short by nearly 215,077 homes, pushing median home prices to historic highs before a gradual cooling began in 2024 and 2025 (KXAN, 2024). This presents a clear need for sustainable, well-sited new housing projects in this region.

This project utilizes GIS-based suitability analysis to identify areas within the Austin-Round Rock-San Marcos MSA that are suitable for the construction of a new sustainable housing development.

Data

This project uses two categories of data: vector datasets representing socioeconomic and infrastructure features and raster datasets representing terrain and land cover. All data were used in ArcGIS Pro and processed within the coordinate system NAD 1983 State Plane Texas Central FIPS 4203 (US Feet), which is the appropriate projection for the Austin-Round Rock-San Marcos MSA. The environmental analysis settings were standardized across all raster operations, with a cell size of 98.4252 feet (equivalent to approximately 30 meters), a processing extent and mask defined by the dissolved study area boundary, and an output coordinate system matching the selected State Plane projection.

Table 1. Summary of datasets used in the suitability analysis.

Dataset	Source	Description	Preparation Steps
Census Tracts (2019)	U.S. Census Bureau	Cartographic boundary shapefiles for all Texas census tracts	Downloaded from census.gov; extracted to Vector folder; reprojected to NAD 1983 State Plane Texas Central FIPS 4203 (US Feet)
Tract_Economy.csv	GEO 5418	Median Household Income (MHI) and population count by census tract for 2019	Columns renamed (VALUE0 → MHI_USD; VALUE1 → POP); joined to census tract shapefile via GEOID field after data type conversion
County	GEO 5418	County boundaries for Texas, including a Metro field identifying the four largest MSAs	Used to select and export Austin-Round Rock-San Marcos MSA counties; dissolved to create StudyArea_Dissolve boundary
Highway	GEO 5418	Major highway network for Texas	Clipped to study area boundary (StudyArea_Dissolve) to produce StudyArea_Highway
Digital Elevation Model (DEM)	GEO 5418	Raster elevation surface for the Austin-Round Rock-San Marcos MSA; cell size 93.53 × 93.53 ft; height values in meters	Used as input for Slope and Aspect geoprocessing tools; no pre-processing required

Dataset	Source	Description	Preparation Steps
Land Cover (NLCD 2011)	GEO 5418	National Land Cover Database raster for the MSA; cell size 98.4252 × 98.4252 ft	Reclassified to binary raster assigning 1 to compatible cover types and 0 to all others using NLCD class codes
Grocery_Stores.csv	GEO 5418	Tabular list of major grocery stores, including longitude and latitude coordinates	Geocoded using Display XY Data tool (GCS WGS 1984); spatially queried to retain only stores within the study area; used as input for Euclidean Distance analysis

Methodology

This project uses the Binary Overlay (Pass/Fail) method of suitability analysis to identify areas within the Austin-Round Rock-San Marcos MSA suitable for a new sustainable housing development. In this method, each location in the study area is evaluated against a set of criteria and assigned a value of 1 (Pass) or 0 (Fail) for each. The seven individual criterion rasters are then multiplied together using the Raster Calculator tool in ArcGIS Pro, producing a final overlay layer in which only cells that pass every single criterion simultaneously receive a value of 1. A cell that fails even one criterion receives a value of 0 in the final output.

Before beginning the criterion analysis, the project environment was configured in ArcGIS Pro to ensure consistent spatial alignment and resolution across all outputs. Under the Geoprocessing Environments panel, the coordinate system was set to NAD 1983 State Plane Texas Central FIPS 4203 (US Feet), and both the current workspace and scratch workspace were pointed to Final_Project.gdb. The processing extent and raster mask were set to StudyArea_Dissolve, and the cell size was set to 98.4252 feet for all raster outputs. These settings remained active for all subsequent analysis steps.

The seven criteria evaluated in this analysis, along with their thresholds, are as follows:

- Criterion 1 — Population Density: Census tracts with a population density ≥ 250 people per square mile

- Criterion 2 — Median Household Income (MHI): Census tracts with an MHI ≥ $50,000
- Criterion 3 — Distance from Highway: Areas located between 0.5 and 10 miles from a highway
- Criterion 4 — Distance from Grocery Store: Areas within 10 miles of a grocery store
- Criterion 5 — Compatible Land Cover: Areas classified as Developed Open Space, Developed Low Intensity, Grassland/Herbaceous, Shrub/Scrub, Barren Land, or Pasture/Hay
- Criterion 6 — Slope: Areas with a terrain slope ≤ 3°
- Criterion 7 — Aspect: Areas with a slope aspect between 112.5° and 247.5° (southeast to southwest facing)

1. Population Density — Population density was calculated by adding two fields to the joined census tract dataset (Join_Tract): AREA_SQMI, using the Calculate Geometry tool set to Square US Survey Miles, and POP_DENS, dividing POP by AREA_SQMI. An attribute query selected all tracts with POP_DENS ≥ 250 people per square mile, which were exported as Polygon_Crit1. A short integer field (RASTER = 1) was added, the layer was converted to a raster using Polygon to Raster and Reclassify was applied to assign 1 to qualifying cells and 0 to NoData, producing Crit1_reclass.

2. Median Household Income — An attribute query on Join_Tract selected all census tracts with MHI_USD ≥ $50,000, exported as Polygon_Crit2. Following the same vector-to-raster workflow as Criterion 1, a RASTER field was created and set to 1, the Polygon to Raster and Reclassify tools were then used to create the final binary output Crit2_reclass.

3. Distance from Highway — The Highway layer was clipped to the study area (StudyArea_Highway), then the Euclidean Distance tool produced a raster of distances in feet from the nearest highway. Reclassify was applied with three classes: 0–2,640 ft (< 0.5 mi) = 0; 2,641–52,800 ft (0.5–10 mi) = 1; > 52,800 ft (> 10 mi) = 0; NoData = 0, producing Crit3_reclass.

4. Distance from Grocery Store — Grocery_Stores.csv was geocoded using Display XY Data (X = Longitude, Y = Latitude, GCS WGS 1984), and a spatial query retained only stores within the study area as StudyArea_Grocery. Euclidean Distance produced a distance raster and Reclassify assigned 1 to cells within 52,800 ft (10 miles) and 0 to all others and NoData, producing Crit4_reclass.

5. Compatible Land Cover — Using the NLCD 2011 raster, Reclassify assigned 1 to the following classes: Developed Open Space (21), Developed Low Intensity (22), Barren Land (31), Shrub/Scrub (52), Grassland/Herbaceous (71), and Pasture/Hay (81). All other classes were assigned 0, producing Crit5_reclass.

6. Slope — The Slope tool was applied to the DEM with output set to Degrees, producing Slope_Crit6. Reclassify assigned 1 to cells with slope 0°–3° and 0 to all cells above 3° and NoData, producing Crit6_reclass.

7. Aspect — The Aspect tool was applied to the DEM, producing Aspect_Crit7 with cell values representing slope-facing direction in degrees clockwise from north (flat areas = −1). Reclassify assigned 1 to values between 112.5° and 247.5° (southeast through southwest) and 0 to all other values and NoData, producing Crit7_reclass.

After generating all seven binary criterion rasters (Crit1_reclass through Crit7_reclass), the Raster Calculator tool was used to multiply all seven layers together, producing the final output

Overlay_AllCrit. Because each raster contains only values of 0 or 1, multiplication ensures that a cell receives a final value of 1 only if it passed all seven criteria simultaneously; a single 0 in any layer leaves a 0 as the result. The Overlay_AllCrit layer therefore identifies, at a resolution of 98.4252 feet per cell, all locations within the Austin-Round Rock-San Marcos MSA that satisfy every criterion for sustainable housing development.

Results

Criterion 1 (Figure 2) maps all census tracts within the study area with a population density of at least 250 people per square mile. Suitable tracts are concentrated along the I-35 corridor, forming a dense north-south band running from Round Rock through Austin and extending southwest toward San Marcos. Smaller clusters of qualifying tracts appear farther east and southeast of Austin, around Bastrop and other growing suburban communities. Substantial portions of the study area in the west, east, and south fall below the density threshold.

Criterion 2 (Figure 3) maps census tracts with MHI of at least $50,000. Much of the study area satisfies this threshold, with only a scattered set of tracts concentrated in and around central Austin and in the San Marcos area falling below it. This pattern reflects generally high-income levels across much of the Austin metro while also indicating pockets of lower-income communities near the urban core and southern portion of the MSA.

Criterion 3 (Figure 4) identifies areas located between 0.5 and 10 miles from a highway, targeting sites accessible to major transportation infrastructure while avoiding highway adjacency. The suitable zone forms a broad corridor running north to south through the center of the study area, closely following the alignment of I-35. A smaller isolated buffer zone also appears in the far southern portion of the MSA near the study area boundary. Large expanses of the western and eastern study area lie beyond 10 miles from any highway and are therefore excluded.

Criterion 4 (Figure 5) maps areas within 10 miles of a major grocery store. Suitable areas form several large overlapping circular buffers centered on grocery store clusters, primarily in the northern and central portions of the study area where store density is highest. The eastern edge of the study area contains one additional isolated buffer around a lone store. The western, far eastern, and southern portions of the study area fall outside the threshold, highlighting the relative lack of grocery access in rural zones.

Criterion 5 (Figure 6) maps areas with land cover types compatible with sustainable housing development. This criterion has the broadest spatial coverage of the seven, with compatible land cover distributed widely across the study area. Gaps in suitable coverage correspond to areas of open water, dense forest, wetlands, and higher-intensity developed land, which are concentrated along river corridors and in the urban cores of Austin and surrounding cities.

Criterion 6 (Figure 7) maps areas where terrain slope is 3° or less, identifying land flat enough for effective residential construction. The overwhelming majority of the study area satisfies this criterion, with unsuitable steep terrain concentrated primarily in the Balcones Escarpment zone in the west-central portion of the MSA. The eastern half of the study area is almost entirely flat and therefore largely suitable under this criterion.

Criterion 7 (Figure 8) maps areas with a slope aspect between 112.5° and 247.5°, corresponding to southeast- through southwest-facing slopes. Suitable aspect values are broadly distributed across the study area in a fine-grained pattern. The pattern closely mirrors the underlying topography, with suitable and unsuitable cells alternating at the scale of individual hillslopes throughout the Balcones Escarpment and the rolling terrain to the east.

The final overlay (Figure 9) shows the result of multiplying all seven binary criterion rasters together, producing Overlay_AllCrit. Only cells that simultaneously passed all seven criteria received a value of 1. The suitable areas form a fragmented but spatially coherent pattern concentrated along the I-35 corridor, running from Round Rock in the north through the Austin metropolitan area and extending southwest toward San Marcos. A smaller cluster of suitable cells also appears in the far southwest of the study area. The eastern and western peripheries of the MSA are largely absent from the results, eliminated primarily by the highway distance, grocery store proximity, and population density criteria.

The attribute table of Overlay_AllCrit contains 532,715 cells with a value of 1, each measuring 98.4252 × 98.4252 feet (9,687.52 sq ft, or approximately 0.000347 square miles). The total suitable area is therefore 532,715 × 0.000347 ≈ 184.85 square miles across the Austin-Round Rock-San Marcos MSA.

Conclusions

This project applied GIS-based suitability analysis to identify approximately 184.85 square miles of land within the Austin-Round Rock-San Marcos MSA suitable for sustainable housing development under seven defined criteria. The Cedar Park and Round Rock areas north of Austin emerged as a particularly strong concentration of suitable land, where high population density, adequate household income, compatible land cover, and proximity to both I-35 and grocery infrastructure converge on relatively flat, south-facing terrain.

Several limitations affect the reliability of these results and should be addressed in any real-world application. First, the socioeconomic data used in this analysis are drawn from 2019 data sets and the land cover data is sourced from the NLCD 2011 data set. Given the high pace of growth in the region, these sets of data are likely to misrepresent current conditions. Second, a binary overlay method treats all criteria as equally weighted, meaning a site that narrowly misses a single threshold is held as having the same suitability as one that fails several criteria by wide margins. Weighted overlays would produce a graduated and more realistic picture of suitability. Third, the grocery store dataset may be incomplete, as it relies on a single provided table rather than a comprehensive, verified database; gaps in store coverage could lead to areas being incorrectly excluded under Criterion 4. Fourth, the DEM used to derive slope and aspect has a cell size of 93.53 feet, which may not capture fine-scale terrain variation accurately enough to classify individual cells reliably; a higher-resolution elevation product would improve the precision of both the Slope and Aspect criteria, particularly along the Balcones Escarpment. Finally, the analysis does not account for land ownership, zoning designations, or other legal constraints that would govern whether any given parcel could actually be developed. Incorporating parcel-level ownership and zoning data would be an essential next step for translating these results into actionable site recommendations.

References

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Appendix

Figure 1. Population Density by Census Tract in Texas (2019)

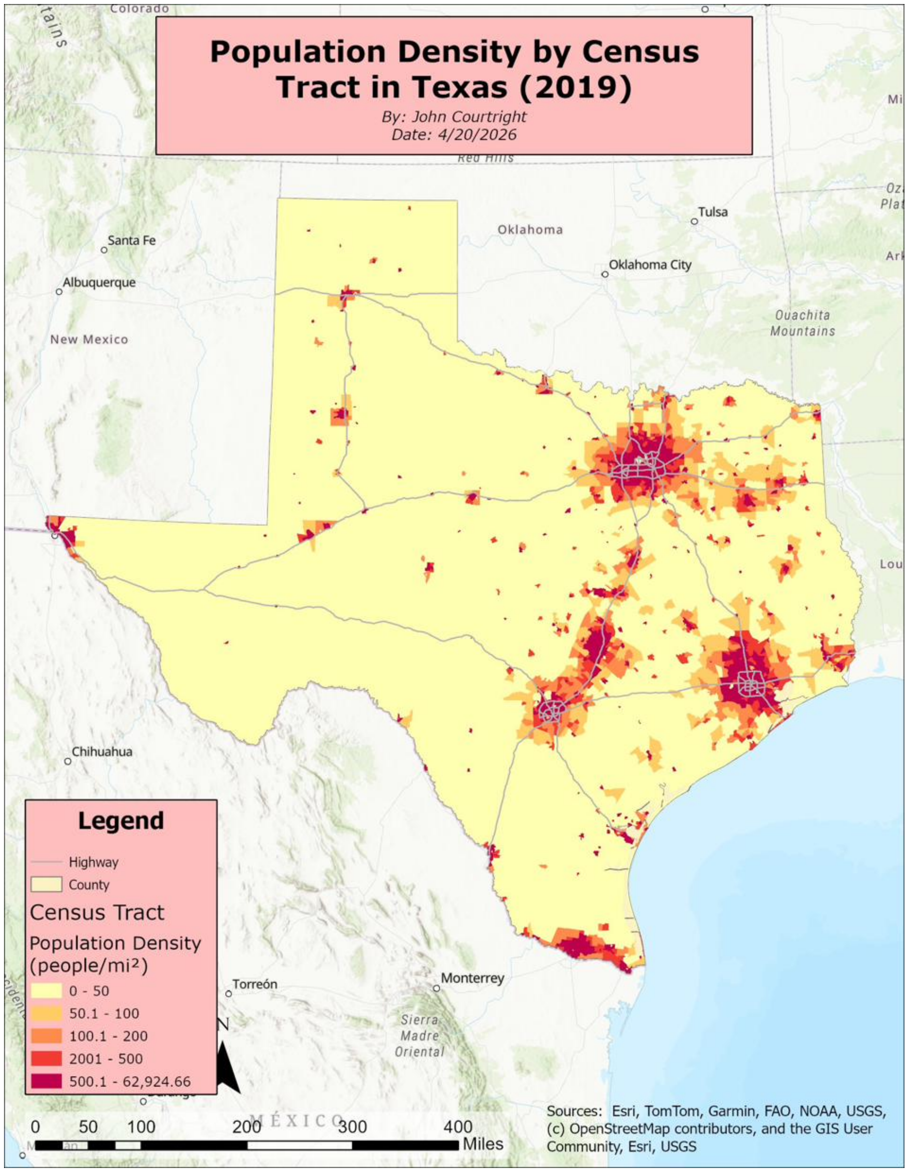

Figure 2. Sustainable Housing Suitability Analysis — Criterion 1: Population Density

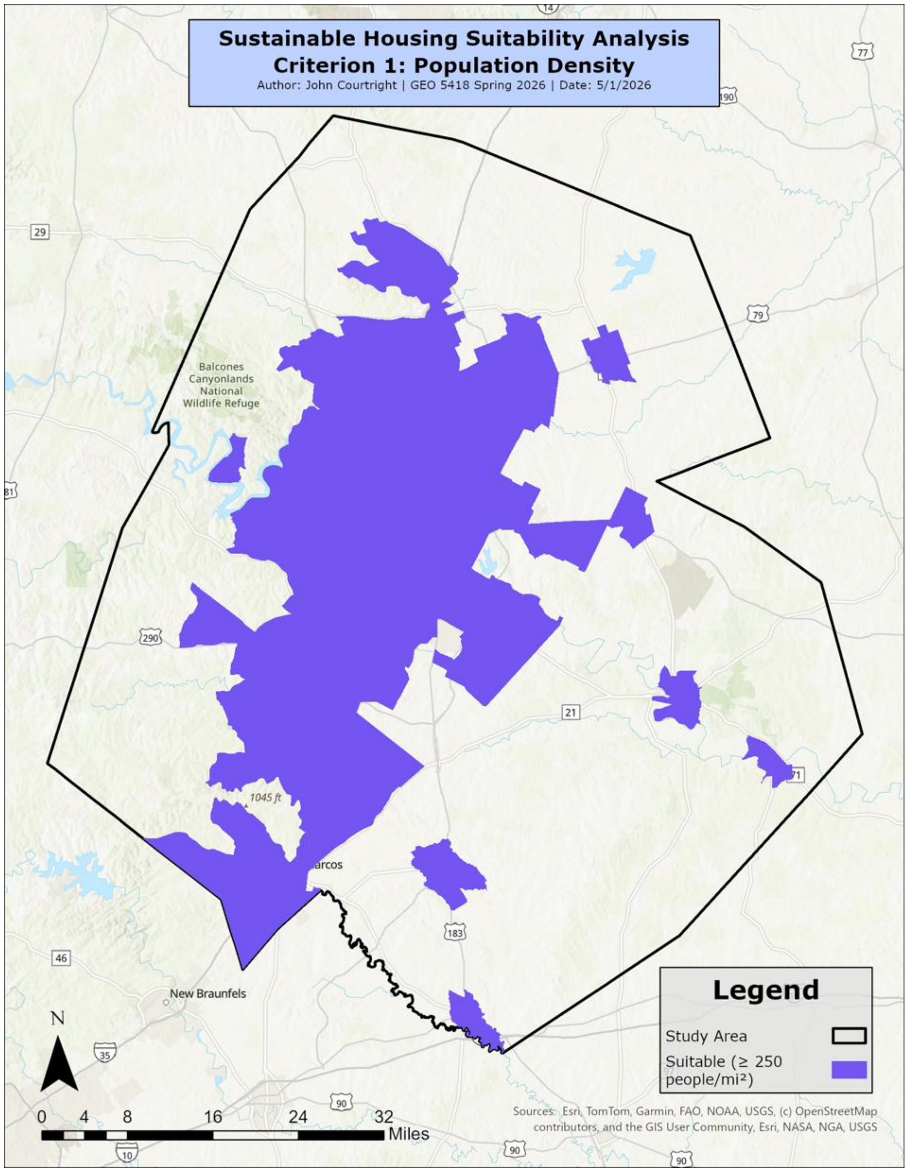

Figure 3. Sustainable Housing Suitability Analysis — Criterion 2: Median Household Income (MHI)

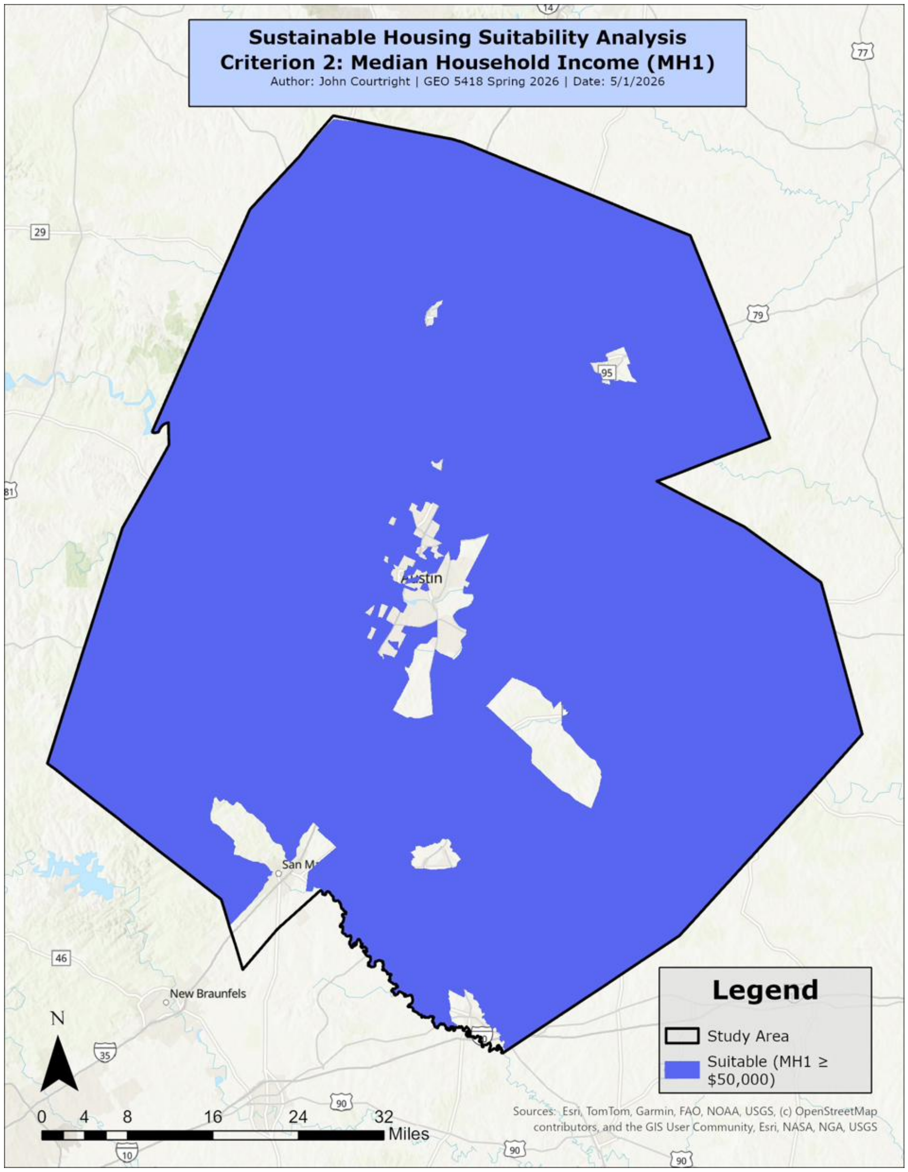

Figure 4. Sustainable Housing Suitability Analysis — Criterion 3: Distance from Highway

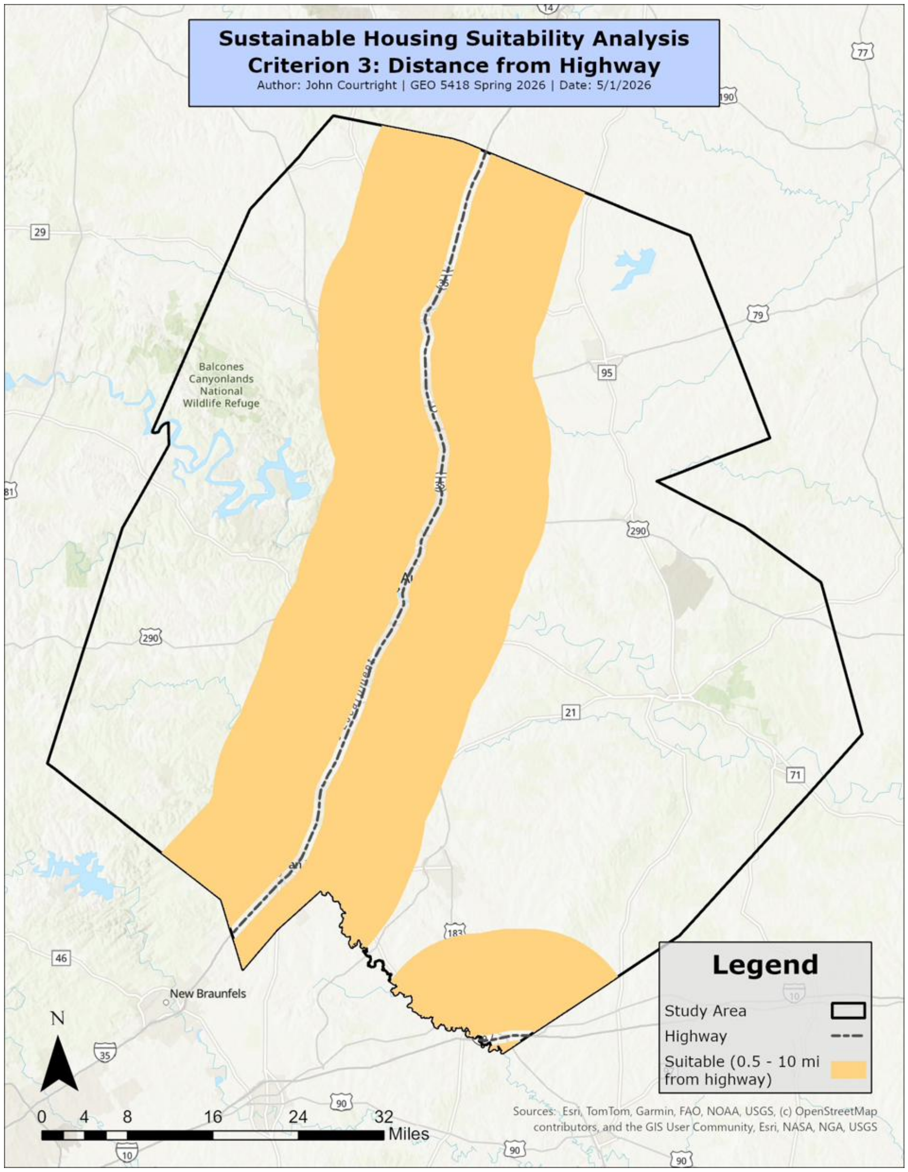

Figure 5. Sustainable Housing Suitability Analysis — Criterion 4: Distance from Grocery Store

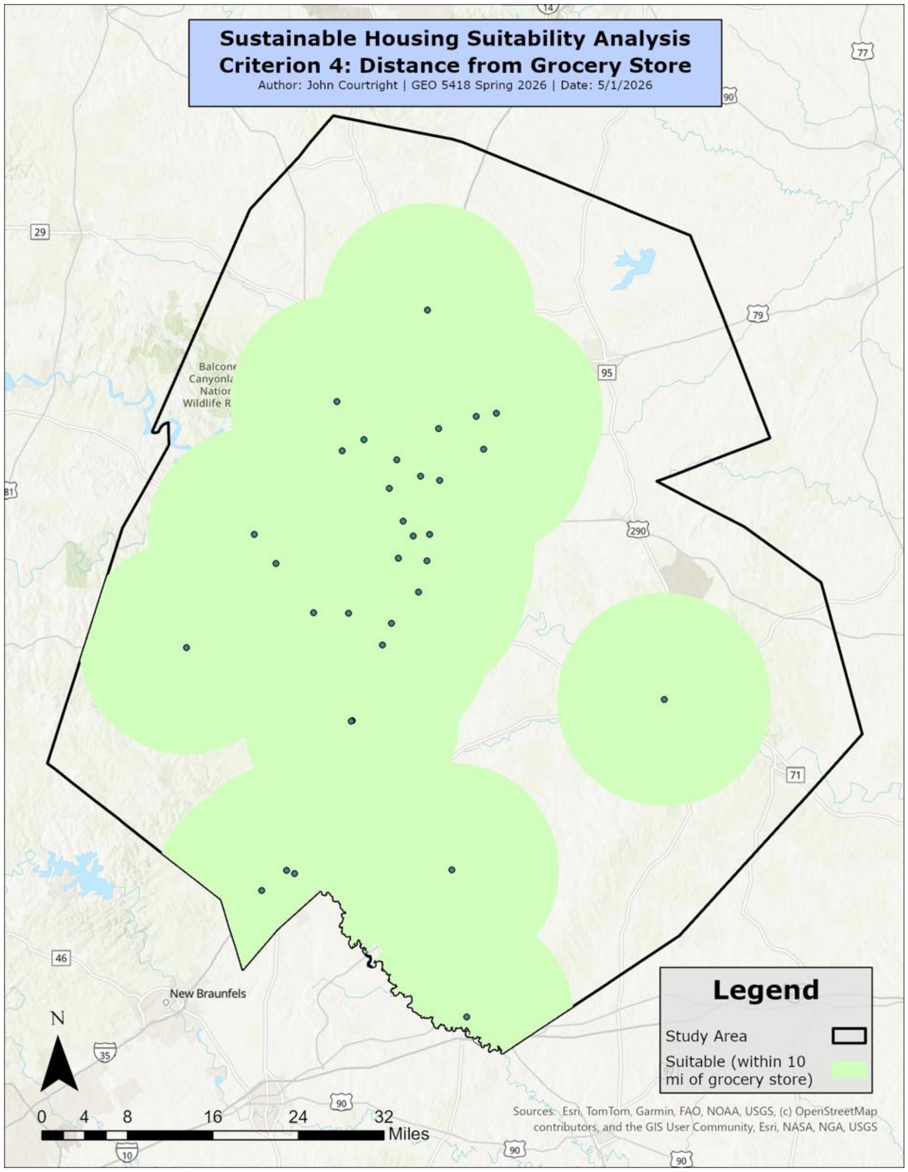

Figure 6. Sustainable Housing Suitability Analysis — Criterion 5: Compatible Land Cover

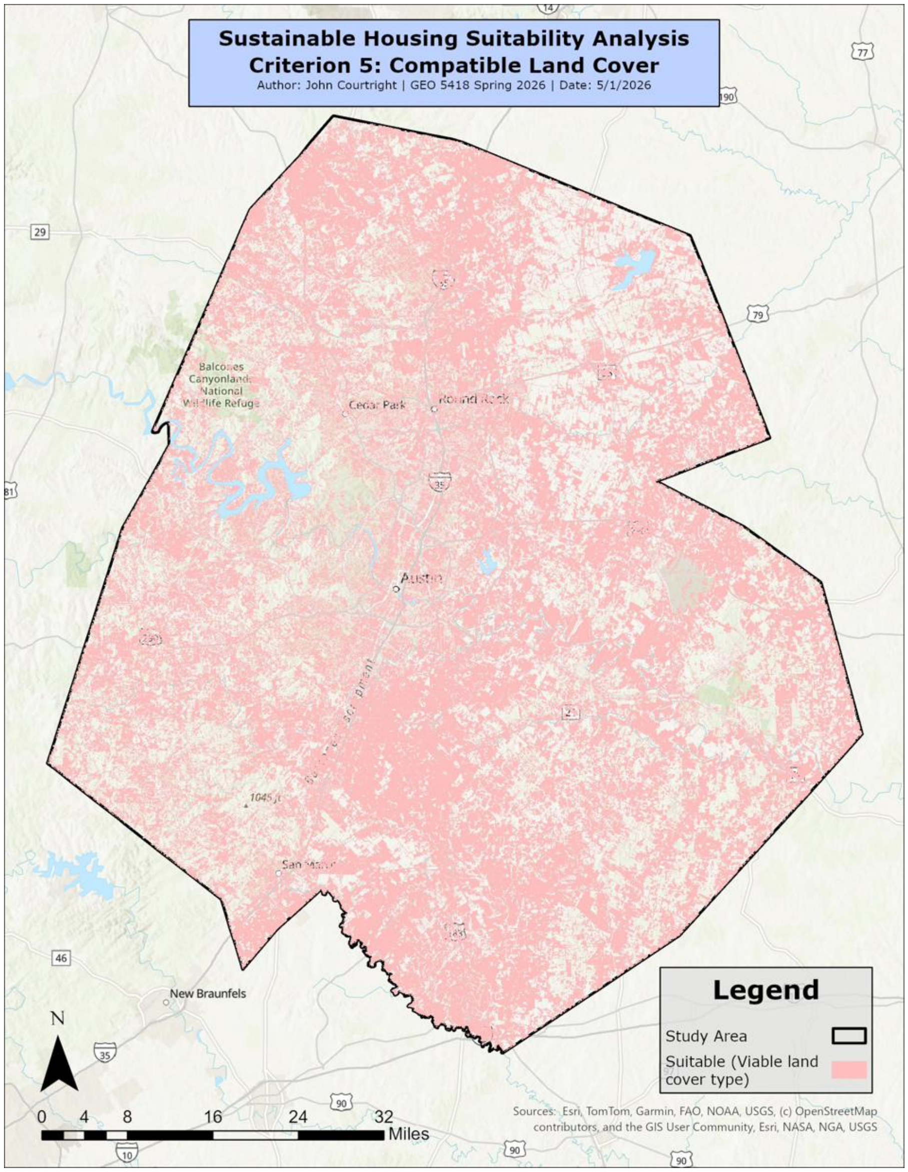

Figure 7. Sustainable Housing Suitability Analysis — Criterion 6: Slope

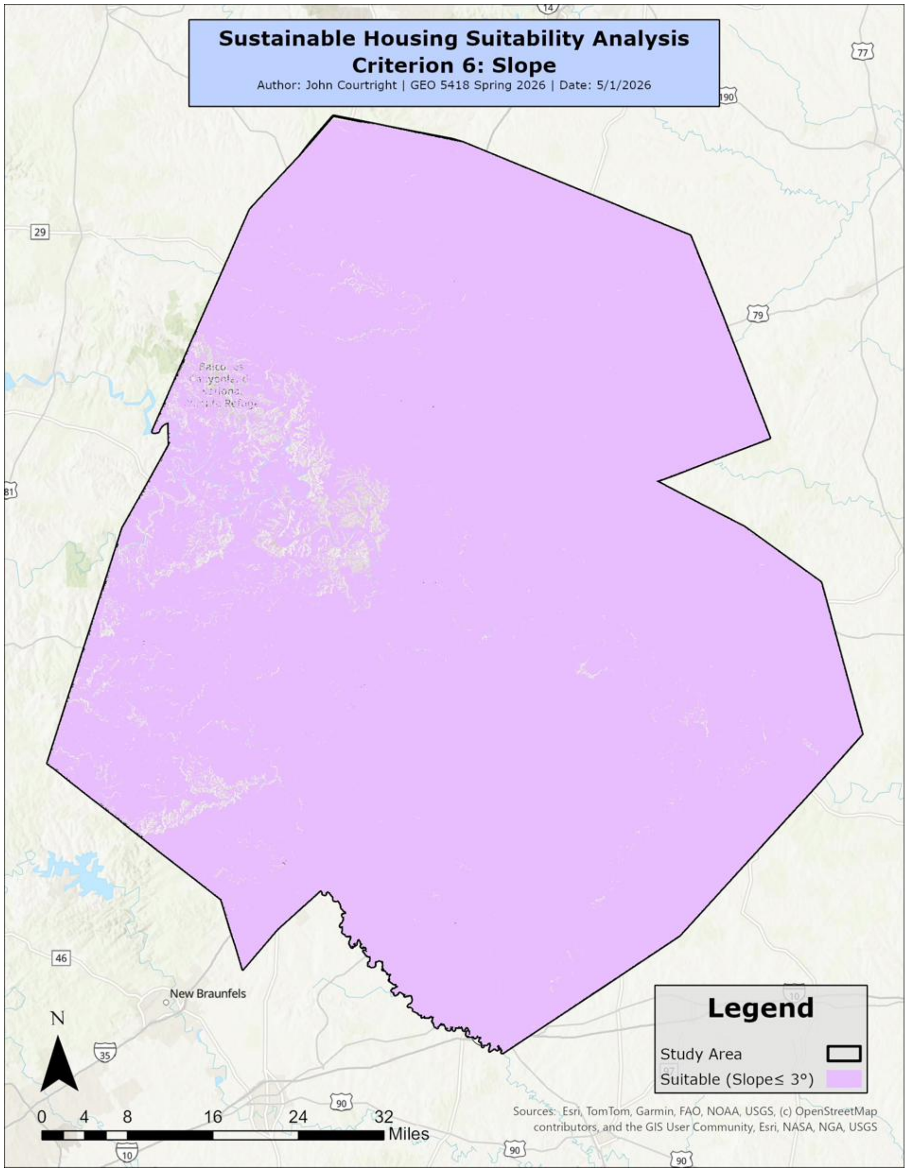

Figure 8. Sustainable Housing Suitability Analysis — Criterion 7: Aspect

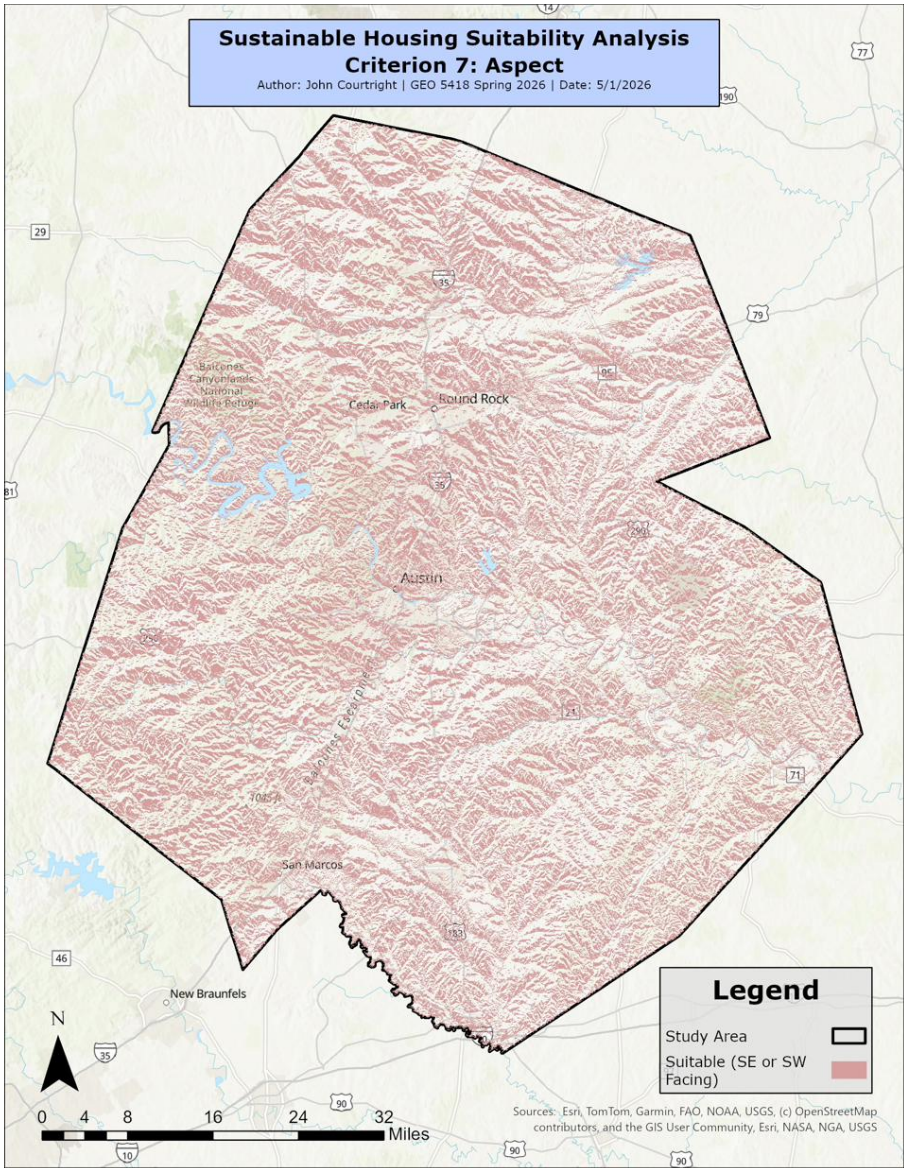

Figure 9. Sustainable Housing Suitability Analysis — Suitable Sites in the Austin-Round Rock-San Marcos MSA